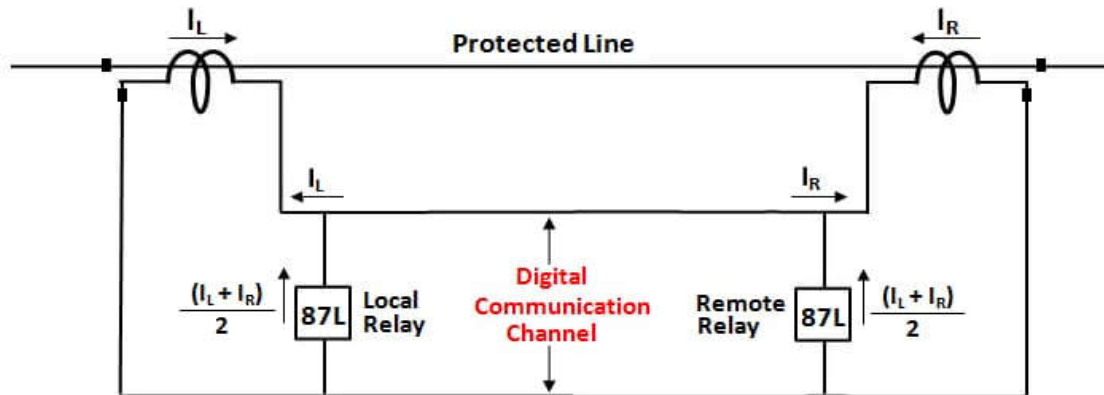


LINE DIFFERENTIAL PROTECTION (87L)

Basic Concept:

Line differential relay protects A transmission line by comparing the current entering & leaving the line section.



Suppose,

Meghnaghat End = Local

Korerhat End = Remote

Differential Pickup :

$$PU = \frac{\text{Rated Current}}{CTR}$$

Suppose Differential pickup current settings is given 0.2

$$\text{Actual pickup} = PU * 0.2$$

From settings = 20% of Rated current

$$CT \text{ Ratio} = 4000/1 \text{ A}$$

Rated current = 1 A (secondary current)

$$\text{Threshold value} = 0.2 I_B$$

$$= 0.2 * 1$$

$$= 0.2 \text{ A}$$

$$I_{diff} > 0.2 = \text{Relay Pickup}$$

Differential Stability:

Local current = I_1

Remote current = I_2

$$I_{\text{diff}} = |I_1 - I_2|$$

$$I_{\text{res}} = \frac{|I_1| + |I_2|}{2}$$

$$I_{\text{diff}} > \text{Slope} * I_{\text{res}} + I_{\text{pickup}} = \text{Relay Trip}$$

$$I_{\text{diff}} < \text{Slope} * I_{\text{res}} + I_{\text{pickup}} = \text{Relay Stable}$$

For Stable Condition,

Local

Remote

$$R < 0$$

$$R < 180$$

$$Y < -120$$

$$Y < 60$$

$$B < 120$$

$$B < -60$$

(180° Phase Difference for normal/External fault)

I_{diff} = Differential Current

I_{res} = Restraining Current

Example,

Meghnaghat Current = 4000 A

Korerhat Current = 3980 A

$$I_{\text{diff}} = 4000 - 3980$$

$$= 20 \text{ A}$$

$$I_{\text{res}} = \frac{4000 + 3980}{2} = 3990 \text{ A}$$

I_{res} = Its Throw Current, not a Fault Current.

$$20 > 0.3 * 3990 + 1.0 = \text{Not Trip} \rightarrow \text{Relay Stable}$$

Differential Unstability:



For Unstable Condition,

Local

Remote

$R < 0$

$R < 0$

$Y < -120$

$Y < -120$

$B < 120$

$B < 120$

(Same Phase Angle for internal fault)

Example,

Meghnaghat Current = 4000 A

Korerhat Current = 1300 A

$$I_{\text{diff}} = |4000 - 1300|$$

$$= 2700 \text{ A}$$

$$I_{\text{res}} = \frac{4000 + 1300}{2} = 2650 \text{ A}$$

$$2700 > 0.3 * 2650 + 1.0$$

= Trip

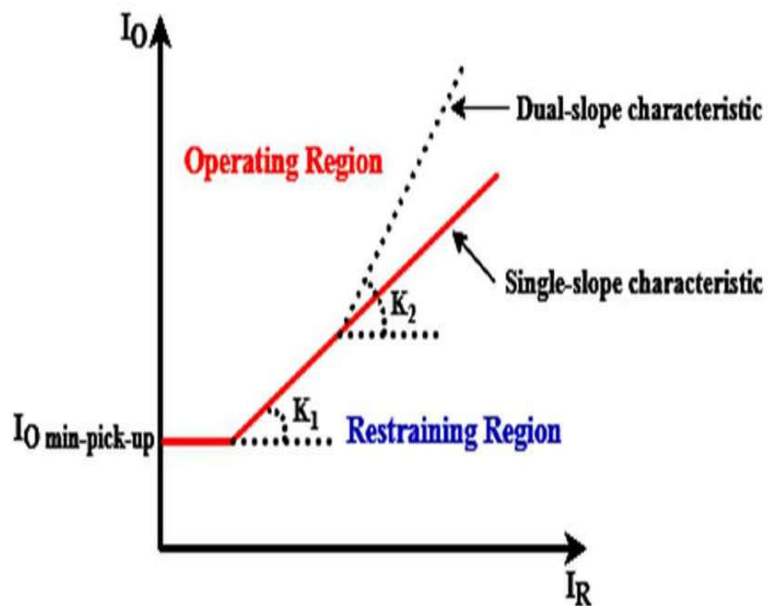
Differential Slope:

Slope = The sensitivity control of the Differential relay

Suppose,

I_1 = Current Entering one side

I_2 = Current Leaving the other side



The slope in a differential relay defines the stable region of the relay. That is how much differential current is allowed before the relay trips. Depending on the level of through current (I_{res}).

Slope 1 = Low I_{res} (load, minor fault) → 20-40%

Slope 2 = High I_{res} (External fault) → 40-80%

For relay

Low I_{res} = Smaller slope → Sensitive

High I_{res} = Higher slope → Stable

$$\text{Slope} = \frac{I_{diff}}{I_{res}} * 100\%$$

87L LINE DIFFERENTIAL FUNCTION TEST

DIFFERENTIAL LOCAL/REMOTE CURRENT MEASUREMENT:

	PHASE	INJECTED CURRENT	INJECTED ANGLE	MEASURED CURRENT MAGNITUDE	REMARK
MEGHNAGHAT	R-N	1A	0	3999 A	LOCAL
	Y-N	1A	-120	3997 A	
	B-N	1A	120	3999 A	
KORERHAT	R-N	1A	180	3996 A	REMOTE
	Y-N	1A	60	3999 A	
	B-N	1A	-60	3999 A	

DIFF PICK-UP TEST:

	PHASE	SET VALUE	MEASURED PICKUP VALUE	TRIP TIME [ms]
MEGHNAGHAT	R-N	0.2IB	0.201A	25.6
	Y-N	0.2IB	0.202A	28.6
	B-N	0.2IB	0.202A	29.3
	RYB	0.2IB	0.201A	30.1
KORERHAT	R-N	0.2IB	0.201A	24.6
	Y-N	0.2IB	0.201A	30.2
	B-N	0.2IB	0.201A	30.4
	RYB	0.2IB	0.201A	27.5

DIFF STABILITY TEST:

	PHASE	INJECTED CURRENT	INJECTED ANGLE	MEASURED RELAY DATA (PRIMARY)	REMARKS
MEGHNAGHAT	R-N	1 A	0	IDIFF R= 17 A	STABLE
	Y-N	1 A	-120	IDIFF Y= 16 A	
	B-N	1 A	120	IDIFF B= 40 A	
KORERHAT	R-N	1 A	180	IRES R= 3999 A	
	Y-N	1 A	60	IRES Y= 3995 A	
	B-N	1 A	-60	IRES B= 3998 A	

DIFF UNSTABILITY TEST:					
	PHASE	INJECTED CURRENT	INJECTED ANGLE	MEASURED RELAY DATA (PRIMARY)	REMARKS
MEGHNAGHAT	R-N	1 A	0	IDIFF R= 7947 A	UNSTABLE
	Y-N	1 A	-120	IDIFF Y= 7992 A	
	B-N	1 A	120	IDIFF B= 7978 A	
KORERHAT	R-N	1 A	0	IRES R= 3999 A	
	Y-N	1 A	-120	IRES Y= 3995 A	
	B-N	1 A	120	IRES B= 3998 A	

DIFF SLOPE TEST-1 (FIRST VALUE) :					
	PHASE	INJECTED CURRENT	INJECTED ANGLE	MEASURED RELAY DATA (PRIMARY)	REMARKS
MEGHNAGHAT	R-N	1.08 A	0	IDIFF R= 877 A	
	Y-N	1.08 A	-120	IDIFF Y= 892 A	
	B-N	1.08 A	120	IDIFF B= 873 A	
KORERHAT	R-N	1.3 A	180	IRES R= 5199 A	
	Y-N	1.3 A	60	IRES Y= 5197 A	
	B-N	1.3 A	-60	IRES B= 5200 A	

DIFF SLOPE TEST-1 (SECOND VALUE) :					
	PHASE	INJECTED CURRENT	INJECTED ANGLE	MEASURED RELAY DATA (PRIMARY)	REMARKS
MEGHNAGHAT	R-N	1.8 A	0	IDIFF R= 2695 A	
	Y-N	1.8 A	-120	IDIFF Y= 2749 A	
	B-N	1.8 A	120	IDIFF B= 2771 A	
KORERHAT	R-N	2.48 A	180	IRES R= 9926 A	
	Y-N	2.48 A	60	IRES Y= 9919 A	
	B-N	2.48 A	-60	IRES B= 9925 A	